

LECTURE 4: CONVENTIONAL ICE NUMERICALS

1. Compute Forces due to drag, rolling resistance and gradient for the following vehicles assuming $\rho = 1.2(\text{kg}/\text{m}^3)$ and $\theta = 8^\circ$. For the three vehicles given in the table, find Aerodynamic drag at velocity v_1 and v_2 ; also find rolling resistance at two velocities.

Vehicle	GVW (kg)	C_D	Area (sqm)	μ	v_1 (kmph)	v_2 (kmph)	Tyre radius (m)
2-wheeler	200	0.9	0.6	0.015	30	80	0.28
3-wheeler	600	0.45	1.6	0.015	30	80	0.2
4-wheeler	1500	0.3	2.5	0.015	30	80	0.3

2. A small electric car has a mass of 1000 kg. The combined coefficient of rolling resistance and aerodynamic drag (C) is 0.25. The frontal area (A) is 2.2 m^2 , air density (ρ) is $1.2 \text{ kg}/\text{m}^3$ and drag coefficient (C_d) is 0.30. The vehicle is accelerating at $1.5 \text{ m}/\text{s}^2$ on a level road at an instantaneous velocity of 60 km/h. Calculate: (a) The force due to aerodynamic drag, rolling resistance, grade resistance and total tractive force (b) Wheel torque and the power required at the wheels to maintain the speed
3. As an example, consider an electric vehicle having a vehicle mass of 1500 kg. Determine the approximate rating of the EV power-train required to accelerate the vehicle from 0 to 100 km/h in 10 seconds, assuming that aerodynamic drag, rolling resistance, and hill-climbing resistance together account for an additional 12% of the required acceleration force. Calculate:
 - a. The total tractive force required.
 - b. The kinetic energy required to accelerate the vehicle from 0 to 100 km/h.
 - c. The time required for acceleration.
 - d. The average power required during acceleration.
 - e. The peak power required at 100 km/h.
4. A car has a mass of 1400 kg, $C_{rr} = 0.015$, $C_d = 0.32$, frontal area $A = 2.1 \text{ m}^2$, air density $\rho = 1.2 \text{ kg}/\text{m}^3$, and drivetrain efficiency $\eta = 0.88$. Calculate:

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- (a) The total tractive force and the power required at the wheels when the car travels at $v = 25\text{ m/s}$ on a level road at steady speed. Also calculate the corresponding engine power.
- (b) If the car accelerates at 1 m/s^2 at the same speed, calculate the additional tractive force and additional engine power required for acceleration.
5. A 3000 kg truck climbs a 6% grade at $v = 10\text{ m/s}$. Given $f_r = 0.02$, $C_d = 1.8\text{ m}^2$ (so use $C_d A$ directly), and $\eta = 0.85$: compute traction power at wheels and engine power. Explain which term (rolling/gradient/aero) is dominant and why. *Air density* = 1.2 kg/m^3